

ACADEMIC DETAILS

Year	Educational Qualification	Institute	Performance
2025	B.Tech. Electrical Engineering	Indian Institute of Technology Kanpur	8.5/10.0
2021	Class XII (BSET)	FIITJEE Junior College, Hyderabad	95.6%
2019	Class X (CBSE)	Kennedy Global School, Hyderabad	97.0%

Conferred with the **Academic Excellence Award** by IIT Kanpur in 2022, given annually to **top 10%** of the batch for excellent academic performance

PROFESSIONAL EXPERIENCE

xTerra Robotics | Strategic and Business Analyst Intern

Nov'22-Jan'23

Objective	• To collaborate with executive team to develop business strategy leveraging market research and quantitative analysis
Methodology	• Prepared a comprehensive report on the business details of 6+ competitors in the Mobility Robotics Quadruped Industry • Leveraged quantitative and qualitative methodologies to perform use-case analysis on 15+ use cases in 7+ industries • Produced a Porter 5 Forces model to provide valuable insights about the business landscape worth about \$75 Million USD
Impact	• Pioneered a formidable foundation for business development , laser-focused on our target market and assisted in fundraising

RESEARCH EXPERIENCE

NSE-ISB Trading Lab, ISB-Hyderabad | ML Researcher

July'23-Present

Collaborating with the subsidiary of Bhartiya Finance Ltd. to develop keys insights in the Indian rural credit system

Objective	• To develop ML-Based Credit underwriting tool based on customer-centric, socio-economic and CSR centric variables
Methodology	• Analysed over 18+ risk assessment metrics of 4.5L customers , incorporated into the model to identify high risk loans • Imported nightlight data of 4 states using Google Earth Engine to include development index further improving the model
Impact	• Tested multiple classification tools and implemented XGBoost with a projected increase of 9.94% in the consumer base • Currently working on making the model region agnostic , to implement in new districts lacking historical lending data

PULSE Laboratory, IIT-Kanpur | Research Intern

Apr'23-July'23

Industry **SURGE Project** funded by Lohum, India's Largest Battery Energy Solutions Provider

Objective	• To develop ML model to predict life cycle of Li-ion cells using early-stage capacity degradation and EIS spectroscopy
Methodology	• Cleaned and Processed 26GB of time-series data consisting of 96,700 cell cycles of 124 commercial cells using MATLAB • Performed various statistics analysis and regularization to prepare 55+ features , ensure minimum multicollinearity • Developed 25+ machine learning models and optimized it using gradient descent algorithms while avoiding overfitting
Impact	• Achieved 93.0% train accuracy and 95.1% test accuracy using Gaussian Exponential model and implemented it on live cells

CASE STUDY COMPETITION

Formula Bharat-2023 Electric | Business Plan Presentation | FSAE Event

July'22-Oct'22

Achieved **First Place** in the event, competing with teams from 20 colleges, aiding IITK to secure overall 1st position

Objective	• To prepare detailed business plan of our electric vehicle by analysing market size and financials and proposing pitch deck
Methodology	• Performed segment-wise analysis to discover new target markets and learned about the potential customers base • Proposed an inflation combat strategy using derivatives, guided by risk management principles ensuring resilience
Impact	• Discovered an untapped target market potentially worth ₹542 crores and pitched the judges asking for ₹10 crores

POSITIONS OF RESPONSIBILITY

Senior Team Member | Team IITK Motorsports, IIT Kanpur

Mar'23-Present

Leadership	• Leading of 2-tier team of 10+ dedicated team members to work towards designing and implement of the software division • Organized written tests and shortlisting interviews for the recruitment of 44 new team members among 140+ applications • Oversaw comprehensive onboarding of new team members and leading brainstorming sessions to improve functioning
Management	• Developed, implementing a timeline, and managing daily to achieve the subsystems targets for Formula Bharat 2024 • Followed Agile method for project planning by preparing Gantt charts to allocate subsystem resources for software division • Prepared market researched optimal budget by ensuring an efficient design for its implementation worth 2.5+ lakhs
Initiatives	• Conducted the introductory lectures of FSEV technicalities for freshers, drawing 450+ attendees to enhance team's image • Instilled a documentation strategy which helps to organise team's technical knowledge and resources for the long-term benefits
Impact	• Secured sponsorship from multiple companies spanning diverse industries and sectors, amount to over +25 lakhs • Coordinated a major collaborative marketing event with Red Bull India , engaging 250+ attendees to promote F1 in campus

KEY PROJECTS

Live Algo Trading Project | Finance and Analytics Club, IIT Kanpur

Oct'22-Nov'22

Agenda	• To generate buy and sell signals on the NIFTY-200 equities with higher integrity and lower risk in intra-day time frame
Approach	• Import OHLC data using Yf module and learned about portfolio optimization and volatility-based trading strategies • Applied mathematical models to design a false-signal algorithm and back-tested the strategy on 2021 minute-wise data
Result	• Attained excessive returns of 6.04% compared to conventional buy and hold strategy over an extended time horizon

OTHER ACHIEVEMENTS

Academics	• Secured an All-India Rank of 1084 and 1466 in JEE ADV and JEE Mains respectively among 1.1 million candidates in 2021 • Listed in the State Top 1% in the Indian Olympiad Qualifier (IOQC) conducted by IAPT in 2021-2022 in state of Telangana • Qualified for Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship award 2021 conducted by IISC, Bangalore
Sports	• Achieved 5th place in National Chess Tournament-2016 at Gujarat in under 1200 category, representing State of Telangana • Achieved Gold medal in Inter-Hall Powerlifting event and Represented IIT-Kanpur in UDGHOSH'22 in 74kgs category
Finance	• Cleared NISM Series XIII: Common Derivatives Exam covering Equity, Currency, Commodity, and Interest-Rate derivative